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Managing logging compensation payment conflicts in Ghana: Understanding actor-empowerment and implications for policy intervention

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ABSTRACT

Forest resources outside permanent forest reserves form a substantial part of the overall natural forest biodiversity resources in Ghana. However, timber exploitation in off-reserve areas has been problematic as loggers and farmers have often been involved in conflicts over payment of crop-damage compensation. The paper examines 81 cases of this conflict which were reconstructed chronologically to investigate patterns of actors' use of power resources and strategies and their effectiveness. The study employed an actor-empowerment conceptualisation of power and used a two-actor game model for the reconstruction of the conflicts. While several strategies and resources were mobilised and employed by the actors, their effectiveness were context-bound. The paper makes a case for areas where actors' conflict capabilities could be built, why mediated bargaining can be more effective for managing such local conflict constellations and the need to interlock actor interests for mutual interaction by institutionalising mediated bargaining.

Introduction

Areas outside permanent forest reserves (off-reserves) constitute an important source for timber. Even though off-reserve harvest has fluctuated over the years, Treue (2001) has demonstrated that more often than not the recorded extraction of timber off-reserves is higher than the recorded on-reserve extraction. On the average, over 50% of total timber harvest in Ghana comes from off-reserve areas (IIED, 1993). However, throughout the high forest zone, crop damage and compensation payment has been observed as a major forest land-use problem in off-reserve areas, often generating conflict between timber contractors and farmers. This situation has created low incentive for farmers to protect trees on their farms (Inkoom, 1999; Amanor, 2002). Yet, since the off-reserve harvest is and has often been high in relation to on-reserve harvest, there is an obvious, albeit indirect, link between low incentives for farmers to protect off-reserve timber trees and the protection of forest reserves. This is because, when off-reserve timber resources are depleted, then the potential scenario is that forest reserves are likely to come under immense pressure for timber. Thus, there is some policy relevance in the contention that the management of compensation-related conflicts is strategic intervention for forest restoration and sustainable forest management (see Marfo, 2007).

However, since conflicts usually involve the pursuit of actors' interest, those actors with access to important resources are often able to draw from them to influence the situation to their advantage. This naturally invites the question of power and as observed by Buckles and Rusnak (1999), power plays a crucial role in natural resource conflicts and conflict management. Mack and Snyder (1957: 242) have long made a thoughtful observation:

> "To the extent that the function of conflict is the clarification and stabilisation of power relations, modes of resolution, which omit or cannot basically affect these relations, are likely to be ineffective".

Thus conflicts offer unique context for understanding power and this understanding can, arguably, stimulate innovations in natural resource conflict management. Marfo (2006) has argued that the logging-damage compensation payment conflict situation provides a suitable context for understanding power-play in two respects. First, it provides a pure case of interpersonal natural resource conflict where power-play could be studied in a two-actor scenario. This is interesting as most natural resource conflicts occur in multi-actor situations but this offers opportunity to compare patterns of actor-behaviour such as reciprocity and tit-for-tat with those observed in other interpersonal social conflict constellations. Second, it involves two major land-use systems, agriculture and

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forestry, which are not only observed as most pervasive conflict issue in off-reserve areas (Asare, 1970; Inkoom, 1999) but also has a scale consideration as it occurs at the community level. The geographical scale within which conflict occurs offer diverse structural opportunities and constraints to the mobilisation of resources and present different scenarios of complexity with regards to size and diversity of actors involved.

However, it has been argued that conventional approaches to the study of conflict have not helped to understand power-play. It has been suggested that a chronological sequencing of conflict episodes can offer a better model for understanding patterns of influence (Marfo, 2006). To explore power-play in compensation payment conflicts, 81 empirical cases of farmer–contractor conflicts was studied.

A brief background to land and tree tenure and logging situation in Ghana is given in the Background and context section. This is followed by a section on Logging and crop-damage compensation payment in off-reserve areas in Ghana, which mainly highlights the legal framework and scholarly observations. This is followed by a section on conceptualising actor-empowerment in conflict using the two-actor game model, which essentially presents a conceptual framework for understanding the role power in conflicts. The methodology used for the empirical investigation and analysis is elaborated in Data and methods while the main findings of the study are presented under the section 'results: patterns in farmer-contractor empowerment process. Finally, the section 'Conflict capabilities in compensation conflicts and implications for intervention' gives the main conclusions of the study vis-à-vis lessons for conflict management interventions and our understanding of power-play in natural resource conflict should also be enhanced.

Background and context

Ghana's forest resources are located in permanent estate, Forest Reserves, and areas outside commonly called off-reserves. In off-reserve areas, timber resources are interspersed in communal lands, farmlands and community settlements. The management of forest resources, both on- and off-reserve is undertaken by the state Forest Service Division (FSD). Although the exact estimation of off-reserve closed forest canopy is sketchy (Inkoom, 1999), the forest resources in the complex mosaic of farmland and secondary forest in these areas are considerable. Treue (2001) has shown that off-reserve timber extraction has always accounted for a substantial share of the total recorded timber extraction in Ghana. There is a substantial stock of trees in traditional cocoa plantations, which require considerable shade tree cover (Inkoom, 1999). Aside patches of communal secondary forests (like sacred groves); most of the off-reserve forest resources are located in farmlands dedicated to food crop cultivation and cash crop plantations.

Land and tree tenure and timber logging rights in Ghana are complex. The general 'customary' law position is that things naturally embedded in, growing on, attached to, flowing through and found on a land are held in trust for the community and administered by the traditional authority (Inkoom, 1999). However, in practice, a distinction between interest in the land itself and interests in things on or attached to the land is often made (Klutse, 1973).

In Ghana, by law (Concessions Act, 1962) all (naturally growing) trees, irrespective of where they occur, are vested in the President in trust for the people. Consequently, the right to control and manage tree resources, including allocation of logging rights, is vested in the state. Farmers have no role in controlling felling on their farms and have no rights to fell timber trees on their farms, though they continue to exercise judgement over which trees to maintain on their farms, during clearing for cultivation for example (Amanor, 1999). Besides, revenue accruing to timber sales, irrespective of source of timber, is shared among the local government body (District Assembly), traditional authorities (Chiefs), Administrator of stool lands (public agency) and the state Forestry Commission. Farmers do not have any share of forest revenue because, by customary conventions, they do not have ownership rights over trees. However, farmers' right of consultation before timber operations has been recently admitted by law:

> "No timber rights shall be granted in respect of lands with farms without the authorisation in writing of the individual, group or owner concerned" (Timber Resource Management Act 1997, s4.2d).

In addition to tree ownership, management, and benefit sharing rights, the general socio-economic and geographical positions of local actors (e.g. farmers) and outsiders (e.g. timber concessionaires) also provide an important context. Most timber firms are located in the cities and towns which are normally far away from the specific areas where timber logging operations are carried out. Besides, public administrative agencies are mostly located in cities and in district capitals. It is not common to find police stations, forestry offices, courts and other public agencies in rural areas in Ghana. Typically, farmers live in villages, and commute daily to their farms and back to their settlement. Farming in Ghana is generally labour-intensive. Poverty is predominant in rural areas accounting for more than 70% of the poor, and in terms of economic activity, poverty is by far highest among crop farmers with about 59% of them living below the poverty line in Ghana (Ghana Statistical and Service, 2000). This income poverty situation has affected levels of education, access to health care and general standard of living. In terms of political setup, most communities are governed by both traditional (chiefs) and local government representatives. None of these local governing structures have formal roles in forest management. Meanwhile, the FSD is virtually not present at the community level due to huge logistical constraint (Asante, 2005). This has created gaps in terms of monitoring of forest operations at the community level. Aside this, local people in general do not have any official role to play in forest management, in spite of the fact that farmers have historically managed natural tree resources on their farms (see Amanor, 1999, 2005). Besides formal roles, farmers are poorly organised, with virtually no political representation in resource management and local decision-making.

Logging and crop-damage compensation payment in off-reserve areas in Ghana

Compensation payment as a result of crop damages of timber operations in farm areas has been one of the most pervasive conflict issues in forest management in Ghana. Long ago, Asare (1970: 10) observed that:

> "The cocoa farmer has developed a more implacable hatred towards the timber contractor than the beasts that thrive on his cocoa fruits and seedlings, now they make sure that during the clearing of the forest every good quality timber tree is destroyed before the contractor menacingly invades his cocoa farm with a caterpillar"

The legal framework covering the issue of compensation for crop damage is quite pluralistic, often complicating the issue. According to the Economic Crop Protection Decree of 1979 (AFRCD[1] 47), it is illegal to fell timber on a cocoa farm, but in practice, this is not

[1] Armed Forces Revolutionary Council Decree.

upheld. Inkoom (1999) noted that concession leases issued after this decree further confuse the issue by stipulating that compensation should be paid for crop damage. The practical approach has been the development of official valuation rates which are determined by Land Valuation Board, but observations are that these are not regularly reviewed and updated.

Several studies have observed that, in many cases, compensations are either not paid or amounts paid are woefully inadequate (Treue, 2001; Amanor, 2002). Inkoom (1999: 105) observes that "while compensation is paid by some companies, this is by no means the rule and in many places, farmers are destroying trees because of inadequate compensation". In connection with this and other forest management issues, timber merchants have traditionally wielded enormous economic and political power and it has been quite difficult to implement forest management practices which in many instances have been contrary to their economic objectives (Kotey et al., 1998; Inkoom, 1999).

This situation has led to destructive coping strategies by farmers (Amanor, 2002; Lambini et al., 2005). For instance Amanor (2002: 316) has observed that 'many farmers destroy timber trees on their land, ostensibly to make way for cultivation, but in reality to deny timber companies any excuse for encroaching on their farms and destroy their crops'.

Conceptualising actor-empowerment in conflict using the two-actor game model

Conflict model

For empirical application of the term 'conflict' in a study to capture the chronological patterns of episodes in natural resource context, Marfo (2006) has argued for the adoption of Glasl's model (1997). The model basically define conflict as occurring when the behaviour of an actor A is perceived/experienced as impairment by actor B due to differences in perception, emotions and interests (Fig. 1).

It is argued that Glasl's model has two fundamental advantages for the empirical study of conflict. First, in the context of natural resource conflicts which usually involve several actors with diverse interests or stakes, Glasl's model helps to take a point of departure from the usual politics of interests (Cochran, 1973) which involves the actions of parties to pursue their interests in the control, use and management of resources. The operational point of delineation of conflict and politics of interest thus involves the experience of impairment of one's goal as a result of the behaviour of another party in the process of pursuing their interests, which may provoke responsive actions to 'manage' the impairment. Second, the use of Glasl's model allows for a theoretical deconstruction of conflict episodes, which may involve multiple actors at several levels (typical in natural resource contexts), into two-actor fields which can be studied independently.

A fundamental point of departure is the notion of conflict capability, to use Glasl's language. The concept is distanced from the conflict avoidance and belligerence schools of thought (Glasl, 1999) and recognises that conflict has both positive and negative capabilities and that conflict management can harness the positive capabilities for social change.

Understanding power using the actor-empowerment framework

The term 'actor-empowerment' was adopted to describe the exercise of power as a process by which conflict actors mobilise resources to gain the capacity to take strategic actions, thus distancing the term from its common usage as an interventionist notion. Clegg (1992) in his frameworks of power, Knights and Vurdubakis (1994) and Coleman (2000), respectively, give a good summary of this actor-oriented insight:

> "Rather than imputing interests, on whatever theoretical basis, the approach favoured here is aligned to perspectives, which seek to demonstrate how networks of interest are actually constituted and reproduced through conscious strategies and unwitting practices constructed by the actors themselves." (Clegg, 1992: 204)

> "We believe that it is more productive to attend to the practices, techniques and methods through which 'power' is rendered operable. By this we mean those procedures, forms of knowledge and modes of rationality that are routinely deployed in attempts to shape the conduct of others" (Knights and Vurdubakis, 1994: 174)

> "Power can be usefully conceptualised as a mutual interaction between the characteristics of a person and the characteristics of a situation, where the person has access to valued resources and uses them to achieve personal, relational, or environmental goals, often through using various strategies of influence" (Coleman, 2000: 113)

Thus, actor-empowerment entails the dimensions of mobilising power resources to take strategic actions of influence (power strategies).

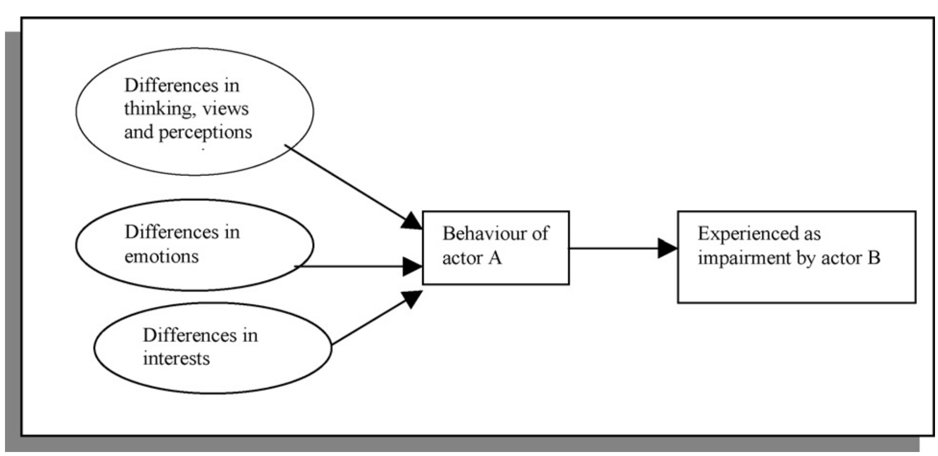

Fig. 1. Glasl's model of conflict (Glasl, 1997).

Table 1
Analytical categorisation of power resources

Operational category	Resource definition	Examples
Socio-economic	Giving access to means of production	Land, money, capital, labour, experts
Social	Access to means of organisation	Networks, group members, politicians, clique
Cultural/orientational	Access to means of orientation	Knowledge, ideology, curse, religion, customs, information, etc.
Institutional	Access to formal state and traditional structures and laws	Law, political and administrative state structure, media, etc.

Source: Marfo, 2006.

Table 2
Possible intentional actions (power strategies) of an impaired actor

Description of conflict	Action to combat impairment (power strategy)
Behaviour of B impairs A	B curtails interactions with A to avoid direct contacts with A in order to sustain the impairing behaviour. This can be called *Avoidance*
Behaviour of A impairs B	B bargains by attempting to convince A to stop or change behaviour. This can be called *Persuasive bargaining* (*persuasion*)
Behaviour of A impairs B	B bargains by attempting to change the setting to constrain A's preferred behaviour. This can be called *Manipulative bargaining* (*manipulation*)
Behaviour of A impairs B	B seeks the support of neutral third party for non-adjudicative intervention to find solution. This can be called *Mediation*
Behaviour of A impairs B	B temporally seeks allies purposely to improve capacity to deal with A's behaviour. This can be called *Coalition-building*
Behaviour of A impairs B	B seeks for avenues to get A to jointly agree to seek non-court adjudication intervention. This can be called *Arbitration*
Behaviour of A impairs B	B seeks for court intervention. This can be called *litigation*
Behaviour of A impairs B	B uses physical force on A to compel A to change his behaviour. This can be called *Force*
Behaviour of A impairs B	It is also possible to imagine B withdrawing or accommodating in order to 'end' the conflict but these are not considered power strategies as they do not have a direct objective to influence A to change his behaviour

Source: Marfo, 2006.

Categorisations of power resources

Power resource can be defined, following Rogers (1974: 1425), as "any attribute, circumstance, or possession that increases the ability of its holder to influence a person or group". Several studies (Rogers, 1974; Korpi, 1985; Giddens, 1982; Joutsenvirta, 1997; Hermens, 1999; O'Riordan and Jordan, 1999; Coleman, 2000; Kurtz, 2001; Scott, 2001; Dugan, 2003a,b,c) have identified a broad range of resources as the basis for the exercise of power. For analytical purposes, a review of these literature shows that they can be categorised into four groups namely institutional, orientational, social and socio-economic. These are explained in Table 1.

Categorisations of power strategies

Power is a highly contested concept (Giddens, 1982; Knights and Morgan, 1991; Haugaard, 1997; Scott, 2001; Galinksky et al., 2003; Spangler, 2003; Arts and Van Tatenhove, 2004) and a clearer delineation of its use, as a root concept of actor-empowerment, is useful. Inspired by several definitions, particularly those of Katz and Kahn (1966) and Wrong (1979), power is conceptualised here as the capacity to exert intentional influence to achieve outcomes where the capacity can be dispositional or mobilised. Thus, in the context of conflict, following Glasl's model, the notion of intentionality is central to operationalising any action as a *powerful* one to the extent that they are undertaken to manage impairing behaviours. Such intentional actions are therefore directed at influencing the behaviour of impairing actors, and hence can be termed 'power strategies'[2]; though the conception of power has also involved unintentional actions (see Giddens, 1982).

Considering hypothetical actors A and B, the possible expected actions to be taken by B when impaired by A can be explained as follows in Table 2.

Notwithstanding the above categorisation of power strategies, in the context of crop-damage compensation conflicts, it can be argued that conflict episodes may end with withdrawal, accommodation or settlement. In relation to the two-actor game model, B is said to have withdrawn if he deliberately stops pursuing actions against 'A' as a result of being less concerned about the impairment. However, B is said to have accommodated when he takes no action against 'A' though he is concerned about the impairment. Aside settlement of compensation, both withdrawal and accommodation are potential end points of conflict episodes. Both withdrawal and accommodation only indicate end points to the conflict but has no relation with payment or settlement.

Understanding actor-empowerment in conflict using a two-actor game model

To study actor-empowerment in natural resource conflict, a chronological sequencing of conflict, using Glasl's model, and the actor-empowerment framework was used to develop a heuristic two-actor game model (Fig. 2).

Hypothetically, assuming that the behaviour of actor A is perceived or experienced by B as impairing, then a conflict situation exists. B responds to managing the impairment (conflict management) taking strategic action (which can be cooperative or competitive) by mobilising power resources (which can be economic, social, orientational and/or institutional); both actions constituting actor-empowerment. Depending on how A perceives B's response, it may compel A to stop his impairing behaviour (inaction) or impair him also, in which case another conflict field is created. A may through actor-empowerment also respond to manage the impairment. The action-response behaviour (episodes) then travel through time ($t_1, t_2 \ldots t_n$) until the conflict is managed. Thus the model allows for the chronological reconstruction of the conflict episodes.

Data and methods

Based on expert advice from the Forestry Commission on the extent of reported cases or occurrence of cases of compensation conflicts, five off-reserve communities were selected. One community, *Wioso*, was selected as a core site where a 100% survey of farmers who had been engaged in compensation conflicts was

[2] Power and influence have for long been distinguished by some Scholars by seeing influence as demonstrated use of power. Influence is a transaction in which one person (or group) acts in such a way as to change the behaviour of an individual (or group) in some intended fashion (Katz and Kahn, 1966).

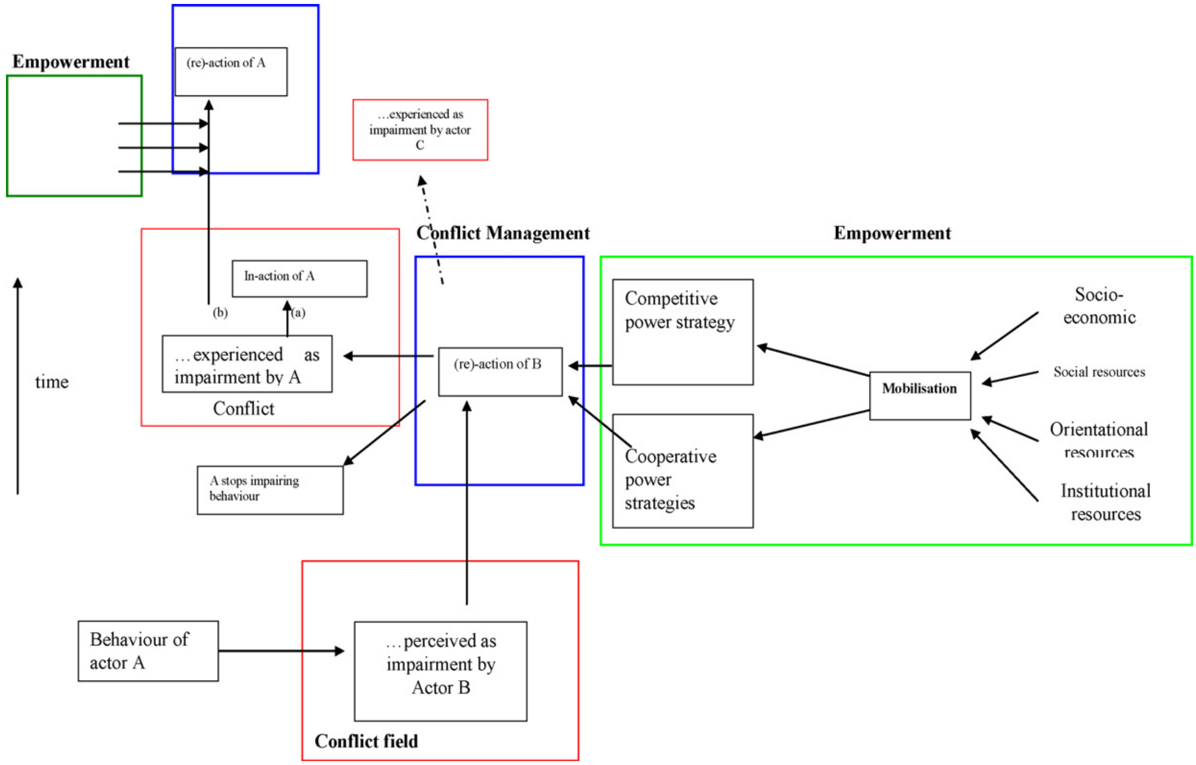

Fig. 2. Actor-empowerment in conflict management using a two-actor game model (source: Marfo, 2006).

selected. In *Wioso*, 63% of farmers were poor, only 11% had a minimum of secondary-level education and only one-third had good knowledge on relevant forestry laws. 46 more farmers were purposefully selected (guided by the qualitative criteria of Bernard, 1995) from the remaining four farming communities to validate patterns observed in the core study area. Table 3 summarises the selection.

The study employed actor narrative of their conflict episodes through semi-structured interview in addition to informant interviews and observations for data collection. In addition to the narratives, data on farmer characteristics such as gender, migration status, income poverty status and educational status were collected. The narratives were used to construct the conflicts chronologically and coded using the categorisations of the power strategies and power resources described in Tables 1 and 2. The constructed dataset captured conflict actors, chronological sequence of power strategies and power resources used by the respective actors. Dataset was analysed for patterns and hypothesis testing using SPSS and Microsoft Excel. An independent peer coding of power strategies and power resources was made. The inter-coder reliability results of about 90% and 92%, respectively were within the recommended acceptable range (Miles and Huberman, 1994).

Table 3
Summary of selection of farmers as respondents for the field interviews

Community	Number of respondent farmers	Sampling type
Wioso (Nkawie)	35	100%
Foase (Nkawie)	28	Purposive selection
Adumasa (Juaso)	4	Purposive selection
Bronikrom (New Edubiase)	2	Purposive selection
Lineso (Bibiani)	12	Purposive selection
Total	81	

Source: Marfo, 2006.

Results: patterns in farmer–contractor empowerment processes

Chronological patterns in conflict episodes

Using the term ‘chronological length or step’ to describe the maximum number of action–reaction incidents that specific episodes ‘lasted’, Fig. 3 compares the chronological sequences of the 81 studied farmer–contractor conflict episodes and their percentage frequencies of occurrence using a time-series graph. On the

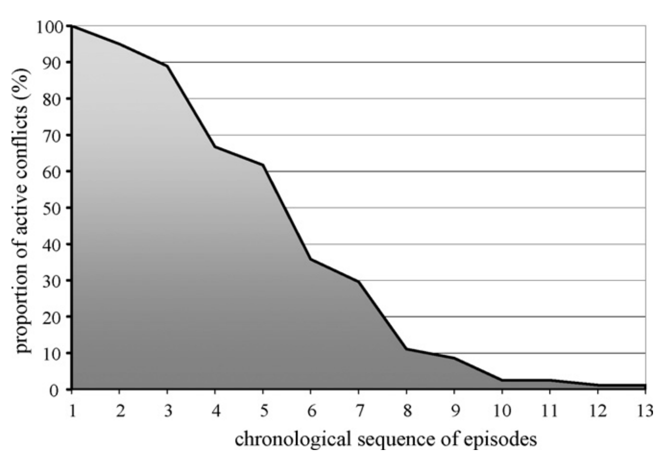

Fig. 3. proportion of the number of conflict actions (active conflict) that travelled through the various chronological sequence steps in the 81 farmer–contractor episodes. In all 409 specific actions at different chronological steps were involved.

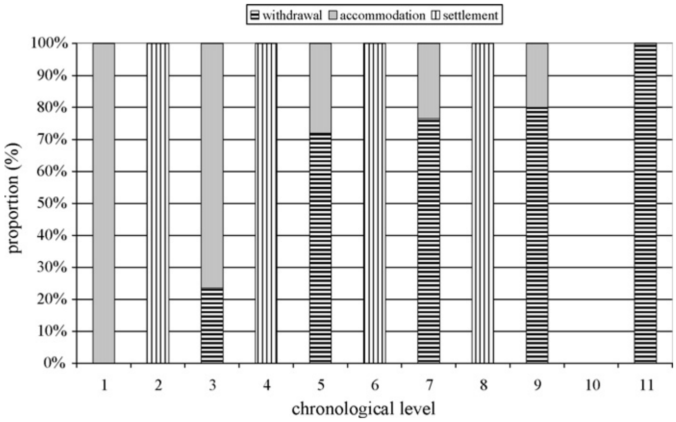

Fig. 4. The number of conflict end points as a percentage proportion of conflict end points at specific chronological sequence steps in the 81 episodes.

average, most conflict episodes 'terminated' at chronological step 5 and were short-lived; less than 10% of travelled beyond chronological step 8. Statistical tests (at $\alpha = 0.01$[3]) showed that farmer characteristics such as gender, income poverty status, educational level and migration status did not significantly affect the chronological length of their conflict interaction with contractors.

Termination of conflict episodes

Following the chronological sequences of the conflicts studied, the proportion of conflicts which were terminated through various end points is summarised in Fig. 4. The conflict episodes terminated at one of three points, settlement of compensation (15%), accommodation (35%) and withdrawal (50%). With the 15% of the conflicts which ended with settlement at steps 2, 4, 6 and 8, there was a progressive decline; 50% at step 2, about 33% at step 4 and about 8% at steps 6 and 8, respectively. Generally, there was a progressive decline in the termination of conflicts through accommodation, from 100% at chronological level 1, 70% at step 3 to 30% at step 9 (Fig. 4).

A reciprocal pattern is seen with conflict episodes progressively ending from step 3 to 9 through farmer withdrawal from the conflict. Beyond this level, all the conflict episodes ended with withdrawal.

Again, statistical tests showed that there is no significant difference between the farmer characteristics such as gender and educational level with regard to their end points in the conflict episodes. Generally, the pattern shows increasing 'hopeless' perception as more farmers were withdrawing as the conflict travels through time. This inquiry was interesting because withdrawal and accommodation connoted different termination status with those accommodating more likely to renew conflict in the future, for example when they encounter contractors again. This is because farmers' narratives suggested that accommodation did not mean that they had permanently withdrawn from the conflict, it only means that some circumstances, such as inability to locate contractor or his agents, have not enabled them to engage in conflict actions.

The long distances between the locations of timber companies and communities, absence of important institutional agencies in

Table 4
Summary of the observed power resource categories employed in the compensation conflict episodes

Main categories	Sub-categories
Orientational	Strategic framing (pleads, petitions, logic) promise, threats and cultural framing (beliefs, trust)
Institutional	Law (e.g. the police), political legitimacy (DCEs, Assemblymen), and administrative legitimacy (FSD officials).
Social	Social network (e.g. family members), corporate network (e.g. native bush managers), traditional network (community chiefs) and geographical-social circumstances such as the relative locations of contractors from communities
Socio-economic	Wealth (e.g. transportation costs, compensation fees)

Source: Marfo, 2006.

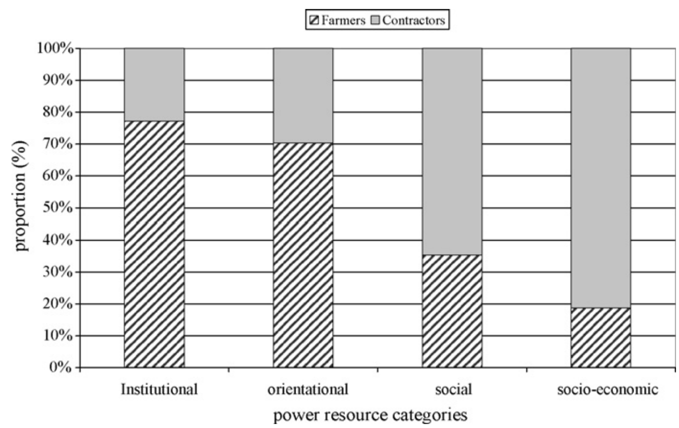

Fig. 5. The proportion of power resource categories used by farmers and contractors in all the studied 81 conflicts (respective number of resources used in all the study cases, $N = 35$, 296, 102 and 32).

rural areas and the general poverty situation collectively provide reasons to account for such observations. Thus, within the active part of the conflict episodes, farmers who do not receive compensation and cannot or are not prepared to pursue the conflict further have only two options, to accommodate hoping that promises may yield some results or withdraw completely. As expected, it is more likely that more farmers may be quite hopeful at the beginning and then withdraw if the situation becomes hopeless as they no longer can encounter contractors. This hypothesis is very much affirmed by the chronological pattern observed as accommodation progressively declined with increasing cases of withdrawal with time.

Patterns in power resources

By counting the frequency of use of the various categories of power resources used as given in Table 4, orientational resources were used in 64% of the cases, 8% for institutional, 21% for social and 7% for socio-economic resources.

Generally, compared to contractors, farmers predominantly used institutional (78%) and orientational (70%) resources. On the other hand, contractors, compared to farmers, dominated the use of social (65%) and socio-economic (about 80%) resources (Fig. 5).

A binomial test[4] to compare the frequencies of the categorical observations of the use of the various power resource groups assuming 50:50 proportions at significant level of 0.05 was carried

[3] As recommended by Bryman and Cramer (1999), for the tests involving two unrelated samples of categorical (gender, migration status) and non-categorical (poverty, educational and knowledge levels) variables, chi-square and Mann–Whitney *U* non-parametric tests were, respectively used.

[4] This test has been recommended for a comparison of the frequency of cases actually found in the two categories of a dichotomous variable, such as gender, with those which are expected on some theoretical basis (Bryman and Cramer, 1999:119).

out. The results show that given a 50:50 chance, farmers are likely to use orientational resources at any given time during their conflict interactions with contractors. For institutional and socio-economic resources, farmers are more likely not to deploy them at any time in their conflict interactions. For social resources, the chances of using them or not were the same.

This observation can be explained using the broader context of the prevailing situation in many rural settlements in Ghana. Obviously, the predominant use of orientational resources can be expected, as, at least everybody can be assumed to have the potential to communicate grievances and attempt to convince or coerce, through verbal communication. Second, at least the 50% probability of using social resource such as family members, social and traditional networks can be explained by the cohesiveness of social structure and interaction in rural Ghanaian settings. With the high poverty incidence in rural settings coupled with the generally low levels of education, it should be understood why many farmers could not use socio-economic and institutional resources, which mutually affect the effective use of each other.

Choice of power resources

Under the general socio-economic, geographical and institutional circumstances surrounding the conflict, it is not surprising that differences in farmer characteristics such as gender, poverty and educational levels may not effectively contribute to differences in conflict capabilities. This is principally because, in many cases, both male and female farmers, poor and non-poor, highly educated and lowly educated and immigrant or native are all faced with the same constraints, which are generally greater than the capabilities these statuses may offer them. For instance, in case of avoidance by contractor, they are all faced with geographical distance constraint, lack of detail information about the contractor such as location of firm and generally less motivated administrative officials to pursue case beyond their office. Besides, even those who have the means to pursue case further may weigh the comparative cost-benefit and may decide to discard the case. Under such practical situations, it may not be resourceful whether a farmer is a native of the community or not, for example.

Chronological sequence of power resources over conflict episodes

For chronological patterns of power resource, orientational and social resources were mainly deployed in the beginning to middle of the conflict episodes (Fig. 6). However, farmers relatively seem to use more of institutional resources than contractors while contractors used more socio-economic resources than farmers. Beyond the use of institutional and socio-economic resources at the latter part of conflict episodes by respective actors, farmers mainly used orientational resources as contractors used social resources (see from chronological levels 10 to 12).

Strategy sequences

Following the chronological sequencing of actor power strategies, it is observed that both farmers and contractors predominantly employed direct persuasive bargaining (over 70% in both cases) at the beginning. However, there was progressive decline in the use of bargaining till it was no longer used by both actors from chronological sequence step 9. Parallel to this, the use of mediation and avoidance, respectively dominated farmer and contractor strategies. This progressively increased from about 15% at chronological step 1 to 100% at step 9 (in the case of farmers) and from about 18% at step 2 to 100% at step 12 (in the case of contractors). The use

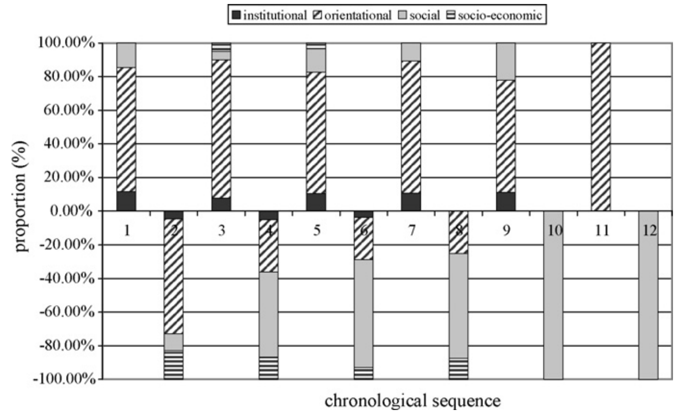

Fig. 6. The number as a percentage proportion of the various power resources categories used by farmers (upper chart) and contractors (lower chart) at specific chronological sequence steps in all the 81 studied conflicts. The total number of resources deployed at chronological steps 1–12 were 96, 88, 79, 61, 58, 28, 28, 8, 9, 2, 2 and 1, respectively.

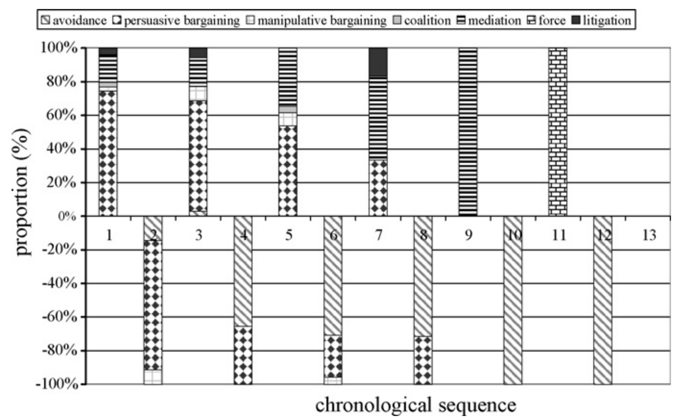

Fig. 7. Number of specific strategies used as a percentage proportion of all strategies used by actors at specific chronological sequence steps in all the episodes. The upper chart is for farmers and lower chart for contractors. The total number of resources deployed at chronological steps 1–12 were 78, 69, 35, 52, 26, 24, 6, 7, 1, 2, 1 and 1, respectively.

of litigation by farmers generally increased with force as the only farmer strategy at level 11.

Due to the predominance of persuasive bargaining, other strategies were regrouped to test the relationship between farmer characteristics and their power strategies. A chi-square test[5] (at $\alpha = 0.5$) gave non-significant results for all characteristics.

Probing into whether escalation or competitive strategies were used in deploying power resources in course of time, Fig. 7 and content of the narratives gave some hints. First, it seems that farmers generally used escalation or more competitive strategies as the conflict episode progressed when the observed pattern of power resources were employed. Farmers started predominantly by using bargaining (step 1) which was followed by a lesser use of bargaining with the frequency of use of mediation and manipulative bargaining increasing. Consequently, persuasive bargaining progressively declined while more competitive strategies like mediation, litigation and coalition-building was increasing till force was deployed at the last level. Generally, the trend of farmers' increased use of addi-

[5] If a comparison of the frequency of cases in a variable in two or more unrelated samples or categories of another variable is needed, chi-square test has been recommended (Bryman and Cramer, 1999: 123).

Table 5
A comparison of power resources used for litigation, bargaining and mediation in episodes that resulted both in settlement and non settlement (resources in *italics* are those which were sometimes effective, hence occurring in both columns)

Farmer power strategy	Resources used to achieve settlement (frequency)	Resources used which did not result in settlement (frequency)
Litigation	Law(3), social network (1)	*Law*(*7*)
Bargaining	Framing(7), cultural framing(1), administrative structures(1)	*Framing*(*78*)
Mediation	Administrative structures(2), traditional network(1), framing(2)	Political structures(6), *Administrative structures*(*4*), *Framing*(*6*), *Social network*(*11*), *Traditional network*(*9*), Geo-social circumstance(2)

Source: Marfo, 2006.

tional strategies, aside persuasive bargaining, such as mediation and litigation suggests increasing attempts to mobilise social and institutional resources. Their description of follow-up actions with contractors or their agents suggested the use of more aggressive and desperate language in their persuasion.

For contractors, a similar conclusion can be drawn due to the progressive decline of persuasive bargaining as a dominant strategy (from step 2 to 8) parallel with the progressive increase in the use of avoidance, which meant increase in competitive or escalation tendencies. The difference between the two seems to lie in the fact that, while farmers increasingly employed resources to maintain contact and interaction with opponents, the contractors progressively pursued avoidance of opportunities for interaction by increasingly capitalising particularly on the geo-social distance between them.

Effectiveness of farmer empowerment

Probing into why some cases were successful (15%) and others not, the first three strategies of farmers were followed since majority of the episodes terminated around chronological sequence step 5. First, it was observed that litigation and mediation were the strategies employed by farmers to achieve full settlement of compensation, even though some cases persuasive bargaining was also effective. However, in a larger number of cases the application of these strategies could not secure settlement. Proportionally, one can say that litigation was the most effective strategy, followed by mediation and then bargaining since the frequency of settlement: non settlement ratios were 0.6, 0.1 and 0.08, respectively. A more qualitative effort through a comparative content analysis of the narratives of the specific cases that resulted in settlement and non-settlement was used. This is because statistical tests on farmer characteristics relation with settlement were non-significant and also power resource-settlement matches did not give strong indications. Table 5 shows that even though same resources were used in the deployment of same strategies in some cases, they were not always useful to achieving settlement. The qualitative analysis of the cases indicates that their effectiveness was context-dependent.

For litigation, the cases that resulted in settlement show that in the pursuit of law, farmer provided evidence such as contractor's car registration number and threatening statements of his agents to the police (case W019). The social circumstances of the farmer as a partially blind person and the use of the friendly relationship between his landlord and the contractor (case W07) were compelling for the police to ensure settlement. Last, when the police arrested the contractor's agent for prosecution (as in case F07), settlement was made. In the other instances of the use of litigation and the law which did not result in settlement, the narratives show that the police attempted to facilitate bargaining, acting more as mediator (as in case F08), the police could not have contacts with the contractor (as in case F10), the contractor was informed of farmer's police complaint (case F14) and farmer agreed with police to social settlement of the case (as in case 020). Thus, the effectiveness of litigation as a power strategy and hence law as a power resource seem to be highly contextual depending on farmer's ability to establish evidence, his willingness to go beyond police reporting, accessibility of the contractor to the police, social circumstances of the farmer and the seriousness and willingness with which the police was determined to enforce the law.

For bargaining, the differences in the narratives were quite fluid and the success of bargaining also seemed to be quite contextual. However, few generalisations could be observed. First, in the instance where prior notice was given to the farmer before logging (cases L10, W022), bargaining resulted in some settlement. Moreover, where specific demands were made and justified in the framing (cases W022, W025, L10), some settlement was made. Another peculiar scenario was in case F06 where prior use of political and administrative structures such as District Assembly and Forestry office to engage the contractor before direct bargaining resulted in compensation payment. In other instances, such as in cases B01 and B02, settlement was made when farmers decided not to challenge or bargain down the offer made by the contractors. Settlement was also made in the only instance where the forestry office (administrative structure) was used by the farmer for the bargaining. In this case, it was complemented by emotionally charged gestures (such as weeping by farmer, case F013). Thus, it seems that strategic framing, cultural framing and the use of forestry administrative structures were more resourceful in farmer bargaining. However, the effectiveness of strategic framing may depend more on farmer's ability to mobilise evidence of offence, use quantitative assessment of crop/property damage and logical argumentations charged with emotional appeal. On the contrary, farmers who did not quantify the damage on crops they have incurred but just stated they had suffered some losses were less effective in getting compensation paid.

For mediation, out of the six cases where structures of administrative legitimacy (forestry service officials) were used by farmers, only two resulted in settlement. Again, out of the 10 cases where farmers used their community chiefs (traditional network) as mediators, only 1 resulted in settlement. For the use of other resources such as political structures (District Assemblies, Unit Committees, Assemblymen) and social network (relatives, contractor's family), no settlement was observed. A content analysis of the narratives of cases where chiefs and forestry officers were used as mediators offers some explanations for the observed effectiveness in ensuring settlement or not. In the case of W028, the chief readily assisted the farmer but in the case of W029, the chief was unwilling to follow-up the matter while in the case of W17, the contractor did not make himself available to the chief. In the cases of W030 and W027, the contractors did not perceive the chief as a legitimate party to mediate in such matters.

With respect to the use of forestry officials, in all the cases that resulted in settlement (cases F01 and F09), the farmers had already engaged the contractors either with direct confrontation or with a chief and had not been 'successful'. In the instances where the involvement of forestry officials did not result in settlement (cases F02, F15, F06), it was observed that the mobilisation was opportunistic, involved lower officials or the forestry office could not summon the contractor to their office. In case F02, the farmer attempted to use the forestry official when he was accompanying the research team to the community. In other instances (case F15), the official was a technical officer and in the case of F06 the contractor refused to respond to the invitation of the forestry office upon several attempts.

In all, the effectiveness of mediation was limited because of the institution with jurisdiction over the settlement of the compensation payment (forestry service) was not 'easily' accessible to farmers. On the contrary, the local traditional and political agencies such as chiefs who were easily accessible did not jurisdiction over the compensation payment issue. Owing to the information such as contractor details and other technical details such as applicable legal rates which were needed for effective mediation, aside the leverage forestry officials (especially higher ranked ones) had on contractors, forestry officials seemed more effective than chiefs and Assemblymen in mediating for compensation settlement. However, the mobilisation of such administrative agencies required money and time since farmers would have to travel to the District office to meet the 'Officer'. There are also possible situations where local leaders could not be effective because their 'neutrality' is tainted as a result of previous transactions with contractors. For example, given the status of chiefs as 'landowners', it is customarily required that any 'stranger' who goes to any community to work must go and 'greet' the leaders especially the chief. During this 'greeting', some donations are normally given for the elders to buy 'drink'. At the same time, some form of soft-negotiation or request is normally made which places some 'social' obligations on the contractor. In such situations, it is practically difficult for the chief and the leaders to be 'neutral' and to help farmers negotiate and enforce compensation payment.

Conflict capabilities in compensation conflicts and implications for interventions

From the results presented above, one can argue that the quality basis of the conflict capabilities of farmers and contractors differ and this observation is crucial for interventions. It can be said that, contractors had high conflict capabilities by effectively using the geo-social circumstances of farmers coupled with their low resource mobilisation capacities to predominantly prevail in the conflict which cumulatively enabled the avoidance strategies to work. Notwithstanding, such avoidance and manipulation are potential breeders of negative coping responses from farmers such as 'unlawful' sale of on-farm trees, deliberate killings and violent confrontations. Therefore, in the sense of broader social and ecological contexts, contractors' conflict capabilities may not be constructive in the long term and in order to manage such conflicts within constructive limits, some deliberate interventions may be needed.

From the patterns observed in the farmer–contractor episodes, it can be argued that conflict management will require two important considerations. First, intervention strategies must keep conflict actors within a virtual geo-social influence boundary in order to ensure constructive interaction. Second strategies should capitalise on 'familiar' power strategies and resources in order to harmonise interventions with actor capabilities and resourcefulness.

Influence boundaries to allow for self-regulated conflict management

A careful study of the chronological pattern of the actor interactions seems to show that that there is a virtual geo-social boundary within which actors predominantly engaged in interactional strategies such as direct persuasion, bargaining and force. We choose to call this, 'influence boundary', and it seems that the closer the relational distance between the actors, the higher the chronological length and the vice versa. In many of the conflicts, the interaction of the actors was conditioned by the presence of the contractors or their agents in the community (relational geographical distance) and when contractors' timber operations (logging, loading and transporting) were ongoing. This business interest creates a social relation between the contractor and the farmers/community. The combination of these realities constitutes a geo-social time boundary within which actors are likely to be able to employ direct interactional influence strategies. Within this time boundary, the chronological intensity will be low if actors are able to reach settlement or influenced the other to adopt yielding actions such as withdrawal. Otherwise, the chronological length may increase leading to long conflict episodes. Lower chronological intensity conflict episodes may be encountered if either actors or one of them live outside the geo-social time influence boundary. These imply that, in order to facilitate actor interactions, they must be kept within an influence boundary by 'interlocking' their interests and keeping their physical accessibility feasible.

The relationship between the chronological intensity of actors' actions within their influence boundary and the observed patterns of power strategies employed in the conflict episodes can be explained using perspectives from the theory of power-dependence relations postulated by Emerson (1964). The theory stipulates that if two persons are unequally dependent on one another for valued outcomes, the less dependent person has a power advantage over the other, and the relation is said to be power imbalanced. Power imbalance is predicted to lead to imbalance in exchange, with the more dependent person giving more than he or she receives (Molm, 1990). In the crop-damage compensation conflict, contractors depend on the cooperation of farmers to carry out activities such as logging and transportation of logs as they occur within the boundaries of farmers' farms and communities. On the other hand, farmers depend on contractors for economic and material resources to compensate the loss of economic resources in the form of damages. The actual power imbalance relationship created in specific cases of farmer–contractor conflict depends on the extent to which the farmer or contractor depends on the other for valued outcomes. However, it seems that in many cases, farmers are more disadvantaged as the symmetry of exchange is often thwarted in favour of contractors who are able to use several means to get their valued outcomes satisfied without depending on farmers. Contractors, aside using unconventional means like night-operations often used promises, for example, to get farmers to allow them to convey their logs or convey logs secretly. By convention then, contractors are able to 'profit' and 'escape' thereby over-thwarting the symmetry of the relationship to create a power imbalance. This implies that farmers are disadvantaged to exercise strategies to keep the balance, resulting in low chronological length. To keep the power balance relationship between the two actors therefore means to keep them within a 'virtual' influence boundary within which the exchange dependence between the two can be sustained. The implications for conflict management intervention is to create mechanisms that interlock the interest or exchange dependence of the two actors before desired social outcomes such as settlement of compensation for damage crops are achieved.

Capitalising on the patterns of actor-empowerment for conflict management

Several concrete implications for pursuing intervention approaches to improving the conflict capability of farmers and for managing the conflict can be made. It seems that alternative dispute resolution (ADR) using approaches like bargaining through mediation (facilitated bargaining) has a high potential for conflict management (CM), owing to the prevalence of mediation and persuasion efforts of farmers in the conflicts. The study shows that chronological intensity of farmer–contractor conflicts can be reduced if collective bargaining can be secured for both farmers and contractors. This implies creating conditions that will enhance farmers' capacity to effectively mobilise social, institutional and orientational resources to pursue persuasion while providing incentive for contractors to bargain by constraining their manipulative tendencies. This means the constraints to collective interaction aided by the geo-social distance between the two must be overcome through strategic institutionalisation of the conflict management mechanism. Therefore, its seems that two important dimensions of policy intervention to secure conflict management mechanism in this context will be capacity building for the actors to access and mobilise relevant power resources and institutional framework within which constructive empowerment strategies can be deployed and their outcomes secured.

From the empirical observation, the strategies that resulted in actual settlement were mainly mediated bargaining and litigation. However, given the low levels of the use of socio-economic resources by farmers coupled with the generally high poverty levels in rural Ghana with average of about 50% incidence below the upper poverty line (Ghana Statistical and Service, 2000), the pursuit of litigation as a strategy may not be generally commendable. Additionally, the existence of inefficient system of administration of justice (Inkoom, 1999), reported corruption in the judiciary and general public mistrust (Marfo, 2006) are further constraints to the use of litigation. For mediation, the claim that it is 'cheap, flexible, adaptable and effective' (Bercovitch, 1989: 286) can be misleading in situations like the farmer–contractor conflict, especially when it is not well institutionalised. The need for institutionalising CM interventions is supported by the fact that, although many direct persuasion or bargaining resulted in some compromised positions, in most cases they never resulted in settlement of compensation mainly because of the avoidance and manipulative strategies of contractors. Therefore, it seems that, self-regulation (leaving the two actors to directly bargain and contest) may not be an effective approach to realising settlement within the shortest possible time.

The argument for mediated bargaining is further strengthened by the observations made by Kressel (2000). He essentially posited that mediation is especially likely to prove useful whenever there are additional obstacles that would make unassisted negotiations likely to fail. We observed that, there are interpersonal, substantive and procedural barriers between farmers and contractors that hindered the use of bargaining. There seemed to be dysfunctional patterns of communication style loaded with anger and emotions which can potentially trigger similar responses, thus creating a bad condition for mutual bargaining or negotiations. Secondly, individual positions on substantive facts usually differed. For instance, how many crops were actually destroyed, the economic value of the destroyed crops and what constitutes 'adequate' compensation were some substantive issues that farmers and contractors had difficulty agreeing on. Lastly, the absence of 'official' or 'recognised' forum for bargaining presents a procedural barrier to effective bargaining. With these barriers prominently existing in farmer–contractor conflict setting at the local level, facilitated negotiation or bargaining seem to be a more useful approach to conflict constellations such as that of the farmer–contractor episodes.

To use mediated bargaining, in the context of such conflict constellations, means fulfilling four essential actor-empowerment conditions; first, building actor capacity for principled bargaining, including orientation of their attitude towards cooperative problem-solving; second, providing bargaining platforms; third, securing mediators that possess the legitimacy as neutral in the eyes of both farmers and contractors and have the capacity to effectively facilitate bargaining and fourth, to ensure legitimisation or legalisation of bargained outcomes.

In terms of actor capacity building for principled negotiation, it seems that both farmers and contractors' capacity to effectively mobilise and deploy resources such as strategic problem framing, provision of evidence and quotation of relevant regulations should be built. The study shows that farmers in particular did not make use of regulations and could not counteract misapplication of administrative permits by contractors. In effect, the implication is that both farmers and contractors need capacity building and orientation towards persuasive communication and bargaining. Second, local leaders (mainly chiefs and unit committee members including Assemblymen), forestry officials and the police were the third party social and institutional agents that were called to intervene. Although, farmers called social network agents to mediate, it is unlikely they will pass the 'neutrality' criterion in the eyes of contractors. It is well documented that traditional and state institutional structures may not be necessarily 'neutral' in natural resource conflict (Doornbos et al., 2000). However, their use, especially by farmers, in the conflict episodes shows that they can be potentially useful. It is this 'weakness' and 'bias tendency' that makes it more useful to consider them as 'mediators' rather than as 'arbiters'.

The study shows that the constraints posed by inadequacy of socio-economic resources and geo-social circumstances may require that the process leading to mediated bargaining should be situated at the community level. Given rural Ghana situation for example where forestry officials and the police may be located outside community, it may tilt the choice of mediators in favour of local structures such as traditional leaders, local government bodies or other institutionalised structures within the community. The challenge obviously will be building the capacity of these structures to effectively perform as mediators. Observations from the study attest to the fact that in many cases, chiefs or local government structures could not effectively mediate conflicts brought to their attention. We further observed that, sometimes chiefs did not even follow the conflicts and where they did, in many cases they only pleaded on behalf of farmers; thus, acting more as allies rather than as mediators to facilitate dialogue and bargaining for settlement. Nevertheless, it was observed that these traditional structures, especially chiefs, still commanded a lot of respect in the eyes of both farmers and contractors which is a potential resource for them to, at least, organise bargaining meetings. Traditionally, chiefs have acted as arbiters in local disputes and this condition potentially require their orientations and capacities to be built on effective mediation.

The potential use of traditional social and institutional structures such as chiefs for mediation should be approached with care, since traditionally, the role of such third parties in conflicts were limited to authoritative and coercive strategies (Kom, 1988). Recent studies in rural governance and natural resource management (Ribot, 1999; Marfo, 2001) give useful precautionary lessons to be observed if customary arbitration should be adopted. First, chiefs are not necessarily neutral and just, and measures to mitigate injustice and partiality should be put in place. This may require building chiefs' capacity for effective arbitration and building trust in them in the eyes of both farmers and timber contractors. Second, this will mean extending forest resource governance to embrace such parties and roles, thereby legitimising their functions. Currently, aside forestry officials, other potential mediators at the local level such as chiefs, Assemblymen and unit committee members do not have any direct role endorsed by law or any administrative control over forest resources.

Finally, securing bargained outcomes or agreements in forms that can be legally tenable as evidence when actors engage in mediation, litigation or arbitration seems to be a crucial intervention. In the study, even when the actors engaged in bargaining to reach agreement on compensation, they were never put in forms that were legally tenable; they only sustained their agreement by verbal repetitions. This gave room for the use of promises by contractors to work effectively and further constrained other efforts to secure payments. Therefore, for mediated bargaining to be effective in managing such local resource-based conflicts like the compensation conflict, a governance mechanism to institutionalise the legalisation of such outcomes as mediated agreements is crucial.

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